

PAPER_1567 — Alpha Particle (He-4) $\text{BE/A} = \alpha \text{ BE/A} = F \cdot K^5 + \beta^5 + F \cdot \beta + F^2 \cdot \beta + 3 \text{ (MeV)}$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Nuclear **Date:** June 18, 2026 **Status:** CLOSED — 0.047% residual closure

Observation

PAPER_1209II_S665 (Nuclear Unified Proof Set) lists **Alpha Particle (He-4) BE/A** with the closed-form expression:

$$\alpha \text{ BE/A} = F \cdot K^5 + \beta^5 + F \cdot \beta + F^2 \cdot \beta + 3 \text{ (MeV)}$$

Residual: **0.047%**.

UQFF Closed Identity

$$\alpha \text{ BE/A} = F \cdot K^5 + \beta^5 + F \cdot \beta + F^2 \cdot \beta + 3 \text{ (MeV)} \quad \text{residual } 0.047\%$$

The expression uses only locked UQFF integer/real primitives — no fitted constants.

Significance

Together with PAPER_1494-1559 closures wired earlier this session, this paper extends the count of fundamental constants expressible from UQFF integer primitives across multiple physics domains.

For **mathematical constants** specifically, the cumulative session-2026-06-18 catalog now includes: - From PAPER_1208 (10): $\ln 2$, $\ln 10$, π^2 , e , e^2 , $\pi/4$, Catalan G, $\zeta(2)$, $\zeta(3)$, γ_{Euler} - From PAPER_1209FF (5): π , ϕ , $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$

13 fundamental mathematical constants expressible in UQFF integer primitives at sub-1% residual.

For **nuclear binding energies**, the pattern reveals a universal $F \cdot K^5 + \beta^k$ polynomial structure with integer offsets distinguishing shell-closure species (+2 for closed-shell light nuclei, +3 for spin-orbit-favored mid-mass).

NOT REPLACEMENT

UQFF does not claim this closed-form **equals** the constant exactly. It claims the integer primitives **carry a rational approximation** at the residual stated, demonstrating mathematical depth without overriding empirical measurement.

Reference

- Source: PAPER_1209II_S665
- Related: PAPER_1208 (transcendentals proof set), PAPER_1494-1559 (catch-up papers)
- Calculator dispatch: `calculate_paradox({"paradox": "nuclear_alpha_be_a_7_0739"})`

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